# Certified Ethical Hacking & AI – Professional

*60-Hour (3 months) Open-Source Ethical Hacking & AI-Assisted Security Program*

| Duration | Level |
|---|---|
| 60 Hours (3 months) | Professional / Practitioner |

## Course Purpose

Practitioner-focused ethical-hacking program covering the complete assessment workflow, common offensive-security domains and responsible AI assistance. Advanced SOC/SIEM engineering, SOAR, AI-agent engineering and AI governance are intentionally reserved for the Expert track.

## Module-wise Duration Summary

| Module | Topic | Hours |
|---|---|---|
| 01 | Cybersecurity & Ethical Hacking Foundations | 3 |
| 02 | Linux Security Lab & Security Tool Fundamentals | 3 |
| 03 | Reconnaissance, Footprinting & OSINT | 4 |
| 04 | Network Scanning & Enumeration | 4 |
| 05 | Vulnerability Assessment Fundamentals | 4 |
| 06 | Web Application Security Fundamentals | 5 |
| 07 | Authentication, Password & Access Security | 3 |
| 08 | Cryptography & Secure Communication | 3 |
| 09 | Network Traffic Analysis & Sniffing | 3 |
| 10 | System Security & Privilege Escalation Concepts | 4 |
| 11 | Wireless Security Fundamentals | 3 |
| 12 | Malware & Threat Analysis Fundamentals | 4 |
| 13 | Social Engineering & Phishing Defense | 2 |
| 14 | Ethical Hacking Assessment & Reporting | 4 |
| 15 | AI-Assisted Ethical Hacking Fundamentals | 3 |
| 16 | AI-Assisted Security Analysis & Automation | 3 |
| 17 | Integrated Ethical Hacking Case Study | 2 |
| 18 | Final Practical Capstone | 3 |
| | **TOTAL** | **60** |

### Module 01: Cybersecurity & Ethical Hacking Foundations — 3 Hours

**Theory:** Cybersecurity & Ethical Hacking Foundations: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Cybersecurity & Ethical Hacking Foundations methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Module 02: Linux Security Lab & Security Tool Fundamentals — 3 Hours

**Theory:** Linux Security Lab & Security Tool Fundamentals: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Linux Security Lab & Security Tool Fundamentals methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, Ubuntu, auditd, UFW/nftables

## Module 03: Reconnaissance, Footprinting & OSINT — 4 Hours

**Theory:** DNS, certificates, technology discovery, metadata, passive/active reconnaissance and OSINT reliability.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Reconnaissance, Footprinting & OSINT methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Amass, dig, nslookup, RDAP/WHOIS

## Module 04: Network Scanning & Enumeration — 4 Hours

**Theory:** TCP/UDP scanning, service/version discovery, NSE and service enumeration.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Network Scanning & Enumeration methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Nmap, Wireshark, isolated lab hosts

## Module 05: Vulnerability Assessment Fundamentals — 4 Hours

**Theory:** CVSS, false positives, validation, remediation and retesting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Vulnerability Assessment Fundamentals methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Greenbone/OpenVAS, Nuclei, controlled lab targets

## Module 06: Web Application Security Fundamentals — 5 Hours

**Theory:** HTTP, sessions, access control, injection, XSS, APIs and security headers.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Web Application Security Fundamentals methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** OWASP ZAP, Burp Community, Juice Shop/WebGoat/DVWA

## Module 07: Authentication, Password & Access Security — 3 Hours

**Theory:** Authentication, Password & Access Security: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Authentication, Password & Access Security methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Module 08: Cryptography & Secure Communication — 3 Hours

**Theory:** Cryptography & Secure Communication: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Cryptography & Secure Communication methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** OpenSSL, GnuPG, CyberChef, Python

## Module 09: Network Traffic Analysis & Sniffing — 3 Hours

**Theory:** Network Traffic Analysis & Sniffing: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Network Traffic Analysis & Sniffing methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Wireshark, tcpdump, Zeek, provided PCAPs

## Module 10: System Security & Privilege Escalation Concepts — 4 Hours

**Theory:** System Security & Privilege Escalation Concepts: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the System Security & Privilege Escalation Concepts methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Module 11: Wireless Security Fundamentals — 3 Hours

**Theory:** Wireless Security Fundamentals: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Wireless Security Fundamentals methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kismet, Aircrack-ng, Wireshark, isolated instructor-owned AP

## Module 12: Malware & Threat Analysis Fundamentals — 4 Hours

**Theory:** Malware & Threat Analysis Fundamentals: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Malware & Threat Analysis Fundamentals methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Ghidra, YARA, ClamAV, safe training samples

## Module 13: Social Engineering & Phishing Defense — 2 Hours

**Theory:** Social Engineering & Phishing Defense: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Social Engineering & Phishing Defense methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Module 14: Ethical Hacking Assessment & Reporting — 4 Hours

**Theory:** Ethical Hacking Assessment & Reporting: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Ethical Hacking Assessment & Reporting methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Module 15: AI-Assisted Ethical Hacking Fundamentals — 3 Hours

**Theory:** LLM-assisted analysis, prompt design, safe query generation and evidence verification.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the AI-Assisted Ethical Hacking Fundamentals methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Ollama, Python, local datasets

## Module 16: AI-Assisted Security Analysis & Automation — 3 Hours

**Theory:** AI-Assisted Security Analysis & Automation: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the AI-Assisted Security Analysis & Automation methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Ollama, Python, local datasets

## Module 17: Integrated Ethical Hacking Case Study — 2 Hours

**Theory:** Integrated Ethical Hacking Case Study: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Integrated Ethical Hacking Case Study methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Module 18: Final Practical Capstone — 3 Hours

**Theory:** Final Practical Capstone: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

**Practical Lab:** Hands-on exercise in an isolated authorized lab: apply the Final Practical Capstone methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

**Tools:** Kali Linux, VirtualBox, Python and controlled lab datasets

## Free / Open-Source Lab Stack

- Kali Linux, Ubuntu and VirtualBox for isolated labs.
- Nmap, Wireshark, OWASP ZAP, Burp Suite Community, Juice Shop/WebGoat/DVWA.
- Greenbone/OpenVAS, Nuclei, John the Ripper, Hashcat, OpenSSL, GnuPG, CyberChef, Python.
- Ghidra, YARA and ClamAV for safe malware-analysis exercises.
- Ollama with a local model for AI exercises; no paid AI API is required.
- Trial products may be used where available, but trial-based software is not described as permanently free.

## Safety & Ethics

- Only test systems you own or have explicit authorization to test.
- Use deliberately vulnerable VMs, synthetic credentials, synthetic logs and instructor-provided PCAPs/samples.
- No public-target scanning, unauthorized credential attacks or destructive activity.
- Wireless labs must use instructor-owned isolated equipment.
- AI-agent exercises use read-only tools unless a controlled action is explicitly approved; validate AI conclusions against evidence.

## Assessment

Knowledge Checks: 15%

Hands-on Labs: 35%

Security Assessment & Reporting: 15%

Integrated Practical Exercises: 15%

AI-Assisted Security: 10%

Final Capstone: 10%

## Student Deliverables

- Rules-of-engagement and lab documentation
- Reconnaissance and network-enumeration reports
- Vulnerability/web/system security reports
- PCAP and malware/IOC analysis where applicable
- Splunk/Sentinel investigation and detection artifacts for Expert learners
- Threat-hunting/ATT&CK artifacts for Expert learners
- AI-assisted analysis or AI-agent project artifacts
- Final capstone report and presentation

## Course Differentiation

This course is the entry professional ethical-hacking track. Learners should complete it before attempting the Expert program if they do not already have equivalent practical experience.

## Reference Alignment

AI-security modules are informed by NIST AI Risk Management Framework guidance and the OWASP GenAI/LLM security project. These references support the curriculum but do not make the course an official certification.

## Important Positioning Note

Independent professional training program. Not official EC-Council CEH courseware and not affiliated with EC-Council, Microsoft, Splunk, OWASP, Kali Linux or MITRE.

## Recommended Delivery

60 hours total. Suggested format: 20 sessions × 3 hours, or 15 sessions × 4 hours, with substantial hands-on lab time.